

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 10: Reduced & Derated Take-Off Thrust (Flex Temp)

Syllabus Source: CAE Oxford (pp. 361-402) | Pass Target: >= 90% AviationExam

1. WHY USE REDUCED THRUST?

Jet engines deteriorate primarily due to high Turbine Gas Temperature (EGT) during take-off at full rated power. When an aircraft is light or runway is long, taking off with reduced thrust saves substantial engine overhaul costs and increases reliability without compromising safety!

2. THE ASSUMED TEMPERATURE METHOD (FLEX TEMP)

Modern turbofans are 'flat-rated' up to a corner point temperature (T_ref, typically ISA + 15°C / ~30°C). Above T_ref, engine thrust drops automatically to protect turbine blades from over-temperature:

- The Flex Principle: The pilot enters an artificial, fictitiously HIGH outside temperature (e.g. Assumed Temp = +52°C, when actual OAT is +20°C). The FADEC commands reduced take-off N1/EPR matching performance on a hot day!
- The 25% Maximum Reduction Limit: REGULATORY LIMIT: Reduced thrust MUST NOT BE LESS than 75% of maximum rated take-off thrust! (Maximum allowable thrust reduction is 25%!).
- Assumed Temp Prohibitions: The Assumed Temperature Method is STRICTLY FORBIDDEN: (1) On contaminated runways (standing water, slush, ice), (2) When antiskid is inoperative, (3) When windshear is reported or suspected!

3. ASSUMED TEMPERATURE VS FIXED DERATED THRUST

Thrust Reduction Method	Certification Status & Minimum Control Speeds	Pilot Operating Authority in Cockpit
Thrust Reduction Method	Certification Status & Minimum Control Speeds	Pilot Operating Authority in Cockpit
Assumed Temperature (Flex)	Certified at full rated thrust. Minimum control speeds (Vmcg, Vmca) are based on FULL RATED THRUST.	Pilot MAY advance thrust levers to full TOGA at any time during take-off!
Fixed Derate (Derated Thrust)	Certified as a completely separate, lower-rated engine (e.g. 24k rating derated to 22k). Vmcg and Vmca are LOWER!	Pilot MUST NOT push levers to full TOGA below Vmca (doing so risks violent loss of directional control)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Assumed / Flexible Temperature

Principles of Flex Takeoff: Step 1: Input fictitious higher temperature (T_flex > OAT) into FMS/MCDU. Step 2: FADEC meters less fuel, delivering reduced thrust to prolong engine turbine life. Regulatory Limits (EASA CAT.POL.A): - Maximum allowable thrust reduction: 25% below full rated takeoff thrust. - STRICTLY FORBIDDEN on contaminated runways (water >3 mm, slush, ice). - STRICTLY FORBIDDEN with anti-skid inoperative or reported windshear.

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Checking Assumed Temperature Feasibility

SCENARIO (Airline Dispatch Problem): An Airbus A320 is operating from a 3,500 m runway: - Actual Outside Air Temperature (OAT): +18°C - Flat-rated corner temperature (T_{ref}): +30°C - Performance tables indicate aircraft can take off safely at current mass up to an assumed temperature of +54°C - Full rated thrust produces 120 kN per engine - At +54°C assumed temperature, thrust output is 93 kN per engine QUESTION: Can this take-off be conducted legally using Assumed Temperature? What is the percentage thrust reduction? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Check Condition 1: Is Assumed Temperature HIGHER than actual OAT? - Assumed Temp (+54°C) > Actual OAT (+18°C) -> PASSED! - (Assumed temperature must ALWAYS be higher than actual OAT and higher than T_{ref}). Step 2: Check Condition 2: Calculate Percentage Thrust Reduction: - Full Rated Thrust = 120 kN - Reduced Thrust at +54°C = 93 kN - Actual Thrust Reduction = 120 kN - 93 kN = 27 kN. - Percentage Reduction = (27 kN / 120 kN) x 100 = 22.5% reduction! Step 3: Compare with the 25% Regulatory Limit: - Regulatory Maximum Reduction = 25.0% - Actual Reduction = 22.5% - Check: 22.5% ≤ 25.0% -> PASSED! The reduction is within the legal 25% limit! Step 4: Check Runway Condition: - Provided the runway is NOT contaminated and antiskid is operational, the takeoff is 100% legal! FINAL ANSWER: Yes! The take-off is legal with Flex Temp +54°C, yielding a 22.5% thrust reduction (below the 25% legal cap).

[!] EXAM TRAP (AVIATIONEXAM) - The Vmcg Difference: Flex vs Derate Trap

- Under ASSUMED TEMPERATURE: Vmcg is based on FULL RATED THRUST (high Vmcg). Moving thrust to full TOGA is safe. - Under FIXED DERATE: Vmcg is based on the REDUCED thrust (low Vmcg). Pushing full TOGA below the full-thrust Vmcg will cause uncontrollable rudder divergence off the runway!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the maximum permissible thrust reduction when using the ASSUMED TEMPERATURE METHOD for

- [A]: 10%
- [B]: Not more than 25% of full rated take-off thrust
- [C]: 50%
- [D]: Unlimited

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CS-25 regulations, thrust reduction using the assumed temperature method is capped at a maximum of 25% below full rated thrust (minimum 75% thrust remaining).

Q2: Under which of the following operational conditions is the use of the ASSUMED TEMPERATURE METHOD

- [A]: At sea level
- [B]: On CONTAMINATED runways (e.g. standing water, slush, or ice)
- [C]: At temperatures below ISA
- [D]: On runways longer than 3,000 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Assumed temperature take-off is banned on contaminated runways due to unpredictable tire friction and the requirement to maintain maximum acceleration and stopping margins.

Q3: What is the fundamental difference between 'Assumed Temperature' (Flex) and 'Derated Thrust' regarding

- [A]: There is no difference
- [B]: Assumed temperature uses Vmcg based on FULL RATED thrust; Derated thrust establishes a certified LOWER Vmcg corresponding to the lower derated thrust rating
- [C]: Derate increases Vmcg
- [D]: Flex has no Vmcg

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Fixed derate is certified at a lower thrust level, allowing lower minimum control speeds (Vmcg and Vmca), whereas assumed temperature retains the higher full-rated Vmcg.